

sxSNF: Similarity Network Fusion-Guided Deep Graph Learning for Single-Cell Multimodal Integration

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Abstract—Accurate identification of cell types from single-cell multimodal data requires integration methods that preserve biological neighborhood structure while reducing modality-specific sparsity and noise. We present sxSNF, a single-cell multimodal integration framework that first constructs modality-specific cell-cell similarity graphs, then performs iterative similarity network fusion (SNF) before self-supervised graph representation learning. This fusion-before-learning design uses the fused graph as a denoised structural prior and further refines it through a masked-edge reconstruction objective with negative sampling. On PBMC-10k and SHARE-seq benchmarks, sxSNF achieved ARI values of 0.694 and 0.589, respectively, and showed higher NMI/AMI values than the evaluated baselines under the same evaluation protocol. On the Chen-2019 SNARE-seq dataset, sxSNF recovered major cortical cell identities, marker-gene programs, and refined oligodendrocyte-lineage subpopulations supported by coordinated RNA expression and ATAC-derived gene activity. These results suggest that combining SNF-based structural denoising with graph learning can improve multimodal single-cell clustering and downstream biological interpretation. The code for sxSNF is available at <https://github.com/labxscut/sxSNF>.

Index Terms—single-cell multimodal integration; similarity network fusion; deep graph learning; cell clustering.

I. INTRODUCTION

SINGLE-CELL multimodal technologies such as scRNA-seq, scATAC-seq, and CITE-seq provide complementary views of cellular identity and regulation [1]–[4]. However, effective integration remains challenging because single-cell multimodal data are highly sparse, heterogeneous, and noisy. Existing computational methods can be broadly grouped into three categories. First, statistical alignment approaches (e.g., CCA, MCA) model cross-modal correspondence through shared low-dimensional structure [5]–[10], but may not fully preserve local topology in sparse scATAC-seq data. Second, deep generative models (e.g., scVI, MultiVI) learn shared latent spaces while modeling technical noise [11]–[14], although the reconstruction objective may not explicitly preserve cell-cell neighborhood connectivity. Recent work such as LIAM (reported as Guanlab-Dengkw in our benchmark)

further extends this family by learning a joint low-dimensional representation for paired multimodal and mosaic integration while addressing complex batch effects through conditional and adversarial training [15]. Third, graph-based approaches (e.g., scGCN, GLUE, scJoint) learn cell embeddings from constructed cell-cell graphs [16]–[18]. Furthermore, graph-based deep learning has shown immense potential in broader biomedical domains, such as medical graph modeling (MGD [19]) and drug-target affinity prediction (MMPD-DTA [20]). However, the performance of these graph-based methods depends strongly on the quality of the initial graph; when modality-specific graphs are noisy or inconsistent, downstream embedding quality can deteriorate.

Similarity Network Fusion (SNF) provides an attractive alternative by iteratively diffusing neighborhood information across modalities to form a consensus similarity structure [21]–[23]. However, most SNF-based pipelines stop at the fused similarity matrix without further deep graph representation learning, limiting their ability to model higher-order nonlinear structure in sparse data.

To address these gaps, we propose sxSNF, a fusion-before-learning framework. Compared with existing GNN-based methods that learn directly from noisy single-modality graphs, sxSNF first applies iterative SNF to construct a densified consensus cell-cell graph [23], [24] and then performs self-supervised deep graph learning on the fused graph. Although this fused graph integrates complementary cross-modal neighborhood information, graphs constructed from sparse high-dimensional single-cell data can still contain low-confidence or spurious connections [25]. Self-supervised graph learning is therefore employed to reconstruct and refine graph structure while learning robust low-dimensional cell embeddings for downstream clustering [26], [27]. This design enables sxSNF to better preserve complementary cross-modal neighborhood structure and to mitigate the effects of high sparsity and inconsistent modality structures.

The principal methodological contributions of this study are summarized as follows:

- 1) We propose a fusion-before-learning strategy in which iterative SNF is used as a graph-level denoising step before self-supervised representation learning, rather than applying graph learning directly to a single noisy modality graph.
- 2) We introduce a masked-edge reconstruction objective with explicit negative sampling and observed-edge preservation to refine the fused graph structure and learn cell embeddings suitable for clustering.
- 3) We evaluate sxSNF through benchmark comparisons, component ablations, parameter-sensitivity analyses,

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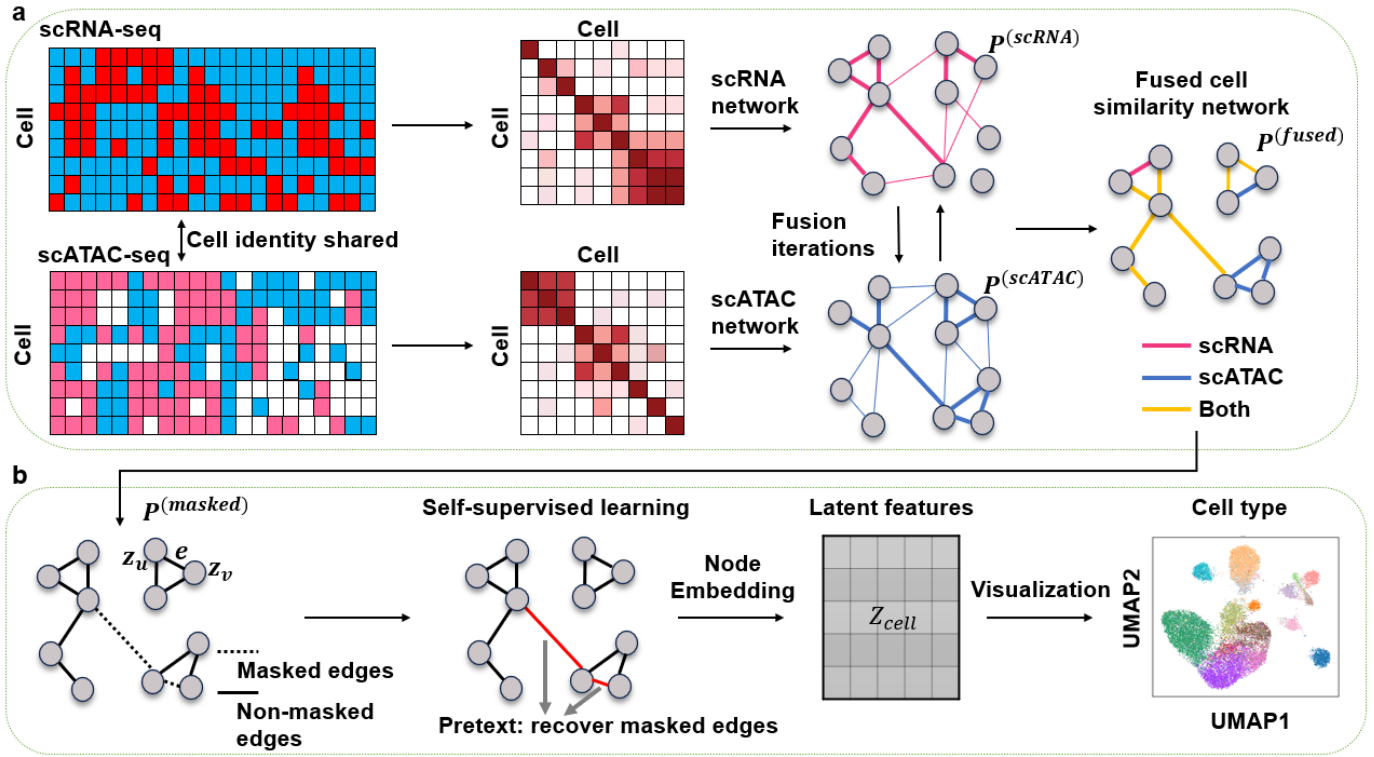


Fig. 1. Schematic of the sxSNF framework. (a) Modality-specific preprocessing generates scRNA-seq and scATAC-seq similarity graphs, followed by iterative SNF cross-diffusion to obtain a consensus cell-cell network. (b) The fused network is used for self-supervised graph learning, in which masked edges are reconstructed to obtain low-dimensional cell embeddings for visualization, clustering, and biological interpretation.

TABLE I
OVERVIEW OF SINGLE-CELL MULTIMODAL DATASETS USED IN THIS STUDY.

Dataset	Species	Tissue	Technology	Cells	Features (RNA/ATAC)	Sparsity (RNA)	Sparsity (Peak)
PBMC-10k	<i>H. sapiens</i>	Peripheral Blood	10x Multiome	11,909	36,601 / 108,377	0.94	0.97
Chen-2019	<i>M. musculus</i>	Cerebral Cortex	SNARE-seq	9,190	28,930 / 241,757	0.97	0.99
SHARE-seq	<i>M. musculus</i>	Skin (Hair Follicle)	SHARE-seq	34,774	17,000 / 370,542	0.97	0.99

integration-quality metrics, and biological validation on cortical multimodal data.

II. METHODS

A. Overview of sxSNF framework

In Fig. 1, we outline the two-stage sxSNF framework for single-cell multi-omics integration. In Fig. 1(a), gene expression (scRNA) and chromatin accessibility (scATAC) data are preprocessed separately, generating modality-specific similarity maps. These are converted into k -nearest-neighbor graphs (pink for scRNA and blue for scATAC), where edge weights reflect pairwise cell similarity. An iterative fusion algorithm aligns the graphs by transferring neighborhood information and reinforcing edges supported by both modalities to produce a unified network. The fused network then undergoes self-supervised learning to produce a low-dimensional embedding (Fig. 1(b)). The core of this learning procedure is a *pretext task* that reconstructs masked edges in the graph. By masking and reconstructing true edges through a self-supervised reconstruction objective, the model learns denoised, structure-preserving

representations. Uniform Manifold Approximation and Projection (UMAP) [28] visualization was employed to project the learned embeddings, revealing distinct cell-type clusters. This workflow enables robust integration, visualization, and analysis of multimodal single-cell data.

B. Datasets and comparative methods

We used three multimodal single-cell datasets in this study. PBMC-10k and SHARE-seq were used as benchmark datasets for quantitative performance comparison, whereas the Chen-2019 SNARE-seq dataset was used for biological validation and interpretability analysis. PBMC-10k is a human peripheral blood 10x Multiome dataset, SHARE-seq is a mouse skin dataset generated using SHARE-seq technology, and Chen-2019 is a mouse cerebral-cortex SNARE-seq dataset. The benchmark datasets were used to evaluate clustering accuracy and integration quality, whereas Chen-2019 was used to assess whether the integrated embedding could recover known cell identities, marker-gene expression patterns, and refined subpopulation structures. For all three datasets, evaluation was performed against the reference cell-type labels provided in

TABLE II
COMPREHENSIVE CLUSTERING PERFORMANCE ON THE PBMC-10k AND SHARE-SEQ BENCHMARK DATASETS (MEAN $\pm$ SD)

Method	PBMC-10k			SHARE-seq		
	ARI	NMI	AMI	ARI	NMI	AMI
Spectral	0.099 $\pm$ 0.012	0.154 $\pm$ 0.015	0.152 $\pm$ 0.014	0.082 $\pm$ 0.010	0.134 $\pm$ 0.011	0.130 $\pm$ 0.010
scMIC	0.114 $\pm$ 0.018	0.301 $\pm$ 0.022	0.291 $\pm$ 0.021	0.153 $\pm$ 0.014	0.324 $\pm$ 0.018	0.315 $\pm$ 0.017
DCCA	0.312 $\pm$ 0.015	0.456 $\pm$ 0.014	0.452 $\pm$ 0.015	0.285 $\pm$ 0.016	0.412 $\pm$ 0.013	0.405 $\pm$ 0.012
Guanlab-Dengkw	0.421 $\pm$ 0.011	0.589 $\pm$ 0.010	0.584 $\pm$ 0.011	0.405 $\pm$ 0.015	0.556 $\pm$ 0.012	0.548 $\pm$ 0.013
SIMBA	0.485 $\pm$ 0.009	0.652 $\pm$ 0.008	0.645 $\pm$ 0.008	0.512 $\pm$ 0.011	0.684 $\pm$ 0.010	0.678 $\pm$ 0.009
scJoint	0.465 $\pm$ 0.010	0.665 $\pm$ 0.011	0.658 $\pm$ 0.010	0.485 $\pm$ 0.012	0.642 $\pm$ 0.013	0.635 $\pm$ 0.012
totalVI	0.472 $\pm$ 0.014	0.678 $\pm$ 0.015	0.670 $\pm$ 0.014	0.490 $\pm$ 0.016	0.660 $\pm$ 0.015	0.652 $\pm$ 0.014
MultiVI	0.488 $\pm$ 0.012	0.695 $\pm$ 0.013	0.688 $\pm$ 0.012	0.505 $\pm$ 0.014	0.675 $\pm$ 0.016	0.668 $\pm$ 0.015
GLUE	0.583 $\pm$ 0.011	0.702 $\pm$ 0.012	0.695 $\pm$ 0.011	0.525 $\pm$ 0.010	0.690 $\pm$ 0.012	0.682 $\pm$ 0.011
Seurat WNN	0.552 $\pm$ 0.008	0.726 $\pm$ 0.009	0.718 $\pm$ 0.008	0.573 $\pm$ 0.009	0.706 $\pm$ 0.011	0.701 $\pm$ 0.010
sxSNF (Ours)	0.694 $\pm$ 0.005	0.833 $\pm$ 0.006	0.832 $\pm$ 0.006	0.589 $\pm$ 0.006	0.768 $\pm$ 0.007	0.761 $\pm$ 0.006

TABLE III
INTEGRATION AND ALIGNMENT METRICS ON THE PBMC-10k AND SHARE-SEQ BENCHMARK DATASETS (MEAN $\pm$ SD)

Method	PBMC-10k			SHARE-seq		
	Batch Mixing	Modality Mixing	Alignment Score	Batch Mixing	Modality Mixing	Alignment Score
Seurat WNN	0.652 $\pm$ 0.014	0.715 $\pm$ 0.011	0.705 $\pm$ 0.012	0.635 $\pm$ 0.016	0.698 $\pm$ 0.014	0.685 $\pm$ 0.015
GLUE	0.685 $\pm$ 0.012	0.742 $\pm$ 0.010	0.730 $\pm$ 0.011	0.665 $\pm$ 0.013	0.725 $\pm$ 0.012	0.710 $\pm$ 0.013
MultiVI	0.670 $\pm$ 0.015	0.720 $\pm$ 0.013	0.715 $\pm$ 0.014	0.645 $\pm$ 0.017	0.705 $\pm$ 0.015	0.692 $\pm$ 0.016
sxSNF (Ours)	0.745 $\pm$ 0.008	0.812 $\pm$ 0.007	0.795 $\pm$ 0.008	0.712 $\pm$ 0.009	0.785 $\pm$ 0.008	0.768 $\pm$ 0.009

the original dataset annotations. The basic statistics of these datasets are summarized in Table I.

For quantitative benchmarking on PBMC-10k and SHARE-seq, we compared sxSNF with representative multi-omics integration baselines, including MultiVI [14], GLUE [17], scJoint [18], Seurat WNN [29], SIMBA [30], scMIC [31], Guanlab-Dengkw [15], DCCA [32], and spectral clustering [33]. We also included totalVI [13] as an auxiliary latent-variable reference. Methods were grouped as deep generative, graph-based, nearest-neighbor fusion, and conventional clustering baselines. For totalVI, which was originally designed for RNA+protein data, we converted the second modality into a matched feature representation for comparison; therefore, its results should be interpreted as an auxiliary baseline rather than a method specifically optimized for RNA+ATAC integration.

C. Experimental protocol for fair comparison

All methods were evaluated using the same filtered cells, the same cell-type annotations, and the same train-free downstream clustering protocol whenever applicable. Unless otherwise recommended by the original implementation, baseline methods were run with their default parameters and repeated under identical random seeds. For each dataset, reported values correspond to the mean and standard deviation across repeated runs. Hyperparameters of sxSNF were selected without using test labels for model fitting; after the default values were determined, they were kept fixed for all repeated runs within the same dataset. For methods that generated latent embeddings rather than discrete labels, the same Leiden clustering and evaluation pipeline were applied to avoid seed-specific or method-specific resolution tuning. The Leiden resolution

was selected at the dataset level according to the expected annotation granularity and then kept fixed for all methods, rather than being optimized separately for individual methods, random seeds, or evaluation metrics. The PBMC-10k benchmark, hyperparameter-sensitivity analysis, and ablation study were evaluated on the same filtered PBMC-10k setting; the full sxSNF result is therefore kept consistent across the corresponding tables. To improve reproducibility, the main parameters used in sxSNF were fixed as follows unless otherwise stated: neighborhood size $k = 20$, SNF iterations $t = 20$, kernel scale $\mu = 0.5$, edge-mask ratio $r_m = 0.2$, one negative edge sampled for each positive masked edge, embedding dimension $d = 64$, Adam optimizer, learning rate 10^{-3} , weight decay 10^{-4} , and 200 training epochs with early stopping based on reconstruction loss.

D. Single-cell data preprocessing

For scRNA-seq data, preprocessing was performed using Scanpy [2]. Genes detected in fewer than 10 cells were removed. Raw counts were library-size normalized, log-transformed, and used to select 2,000 highly variable genes. The selected expression matrix was scaled and projected into a low-dimensional PCA space before graph construction.

For scATAC-seq data, peaks detected in fewer than 100 cells were removed. The peak matrix was binarized and transformed using a TF-IDF-like weighting followed by singular value decomposition (SVD/LSI). The number of retained components was selected from the scree-plot elbow, and peaks with high absolute loadings across these components were used as variable chromatin-accessibility features [34]. For the Chen-2019 biological analysis, ATAC-derived gene activity was computed by aggregating accessibility signals near gene

bodies and proximal regulatory regions, followed by the same normalization strategy used for group-level heatmaps.

E. Similarity Network Fusion

Our similarity network fusion (SNF) approach first constructs a modality-specific network for each data modality. Based on the principle of neighborhood information sharing, the modality-specific networks are then iteratively fused into a consensus network. The fused network captures both shared and complementary information across modalities, thereby reducing modality-specific noise and batch effects.

1) *Similarity network for each modality*: For each modality of the n cells, we formulate a graph $G = (V, E)$, where the vertex set $V = \{v_1, v_2, \dots, v_n\}$ represents cells, and edges E are weighted by pairwise cell similarities. These weights are encoded in a symmetric similarity matrix $W \in \mathbb{R}^{n \times n}$, with $W(i, j)$ defined using a scaled exponential kernel:

$$W(i, j) = \exp \left(-\frac{\rho^2(v_i, v_j)}{\mu \cdot \varepsilon_{i,j}} \right) \quad (1)$$

Here, $\rho(v_i, v_j)$ denotes the distance between cells v_i and v_j , and $\mu \in [0.3, 0.8]$ is a tunable hyperparameter. The scaling factor $\varepsilon_{i,j}$ mitigates local density variations and is computed as:

$$\varepsilon_{i,j} = \frac{\bar{\rho}(v_i, N_i) + \bar{\rho}(v_j, N_j) + \rho(v_i, v_j)}{3}, \quad (2)$$

where

$$\bar{\rho}(v_i, N_i) = \frac{1}{|N_i|} \sum_{v_k \in N_i} \rho(v_i, v_k). \quad (3)$$

Let N_i denote the neighbor set of cell v_i obtained via K -nearest neighbors (KNN). This local scale definition avoids reusing fixed indices as summation indices and follows the SNF intuition that the bandwidth should adapt to the local neighborhood density of each cell.

To enhance numerical stability during normalization, we avoid standard row-normalization, which can be sensitive to diagonal dominance. Instead, we apply the following robust normalization:

$$P(i, j) = \begin{cases} \frac{W(i, j)}{2 \sum_{k \neq i} W(i, k)}, & j \neq i \\ 0.5, & j = i \end{cases} \quad (4)$$

We further refine the local affinity structure using the sparse requirement:

$$S(i, j) = \begin{cases} \frac{W(i, j)}{\sum_{k \in N_i} W(i, k)}, & j \in N_i \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

This construction ensures that similarities between non-adjacent cells are explicitly set to zero.

In our study, we adopt the above framework for the raw modality graph embedding. We also investigate the impact of distance metrics:

- **Cosine Distance**. We test ρ with cosine distance. Cosine distance focuses on the directional agreement between

data points. This makes it more effective for high-dimensional data, where the overall pattern (direction) is more important than the magnitude of individual features.

- **Chi-Square Distance**. For discrete or binned data, such as single-cell RNA sequencing count matrices, we test chi-square distance. This metric is particularly suited for frequency data, as it measures the discrepancy between two distributions while accounting for their expected proportions.

Cosine distance was used as the default distance measure for the main experiments because it was more stable in sparse high-dimensional feature spaces; chi-square distance was retained for sensitivity analysis. Unless otherwise specified, $k = 20$, $t = 20$, and $\mu = 0.5$ were used as the default SNF configuration.

2) *Similarity network fusion across modalities*: Each modality-specific similarity network is characterized by two matrices: the global state matrix P (Eq. 4), which captures full pairwise similarities between all cells, and the sparse kernel matrix S (Eq. 5), which encodes only the local similarities to the k -nearest neighbors. In the subsequent fusion process, P serves as the initial state, while S acts as the diffusion kernel, effectively preserving local graph structure and enhancing computational efficiency.

Following the preceding graph construction steps, we obtain similarity matrices $W^{(v)}$, state matrices $P^{(v)}$, and kernel matrices $S^{(v)}$ for each modality $v = \{\text{scRNA}, \text{scATAC}\}$. The core of Similarity Network Fusion (SNF) lies in the iterative cross-diffusion of state matrices:

$$P_{t+1}^{(\text{scRNA})} = S^{(\text{scRNA})} \cdot P_t^{(\text{scATAC})} \cdot \left(S^{(\text{scRNA})} \right)^T \quad (6)$$

$$P_{t+1}^{(\text{scATAC})} = S^{(\text{scATAC})} \cdot P_t^{(\text{scRNA})} \cdot \left(S^{(\text{scATAC})} \right)^T \quad (7)$$

Here, $P_{t+1}^{(\text{scRNA})}$ and $P_{t+1}^{(\text{scATAC})}$ denote the updated state matrices after t iterations for scRNA and scATAC modalities, respectively. This update scheme creates two mutually reinforcing diffusion processes. After t iterations, the consensus matrix $P^{(\text{fused})}$ (see Fig. 1(a)) is computed as:

$$P^{(\text{fused})} = \frac{P_{t+1}^{(\text{scRNA})} + P_{t+1}^{(\text{scATAC})}}{2} \quad (8)$$

Without loss of generalizability, the update rule of $P^{(\text{scRNA})}$ can be expressed at the element level as:

$$P_{t+1}^{(\text{scRNA})}(i, j) = \sum_{k \in N_i} \sum_{l \in N_j} S^{(\text{scRNA})}(i, k) \cdot S^{(\text{scRNA})}(j, l) \cdot P_t^{(\text{scATAC})}(k, l) \quad (9)$$

This formulation reveals the intuition behind SNF: if two cells v_i and v_j share common neighbors across both modalities, they are more likely to belong to the same cluster. The process allows similarity information to propagate across modalities—even if two cells are not directly similar in one modality, their similarity can be inferred through the structural context provided by the other ($P^{(\text{scATAC})}$ updates similarly).

The algorithm naturally generalizes to $m > 2$ modalities. The iterative update rule becomes:

$$P_{t+1}^{(v)} = S^{(v)} \cdot \left(\frac{\sum_{u \neq v} P_t^{(u)}}{m-1} \right) \cdot \left(S^{(v)} \right)^T, \quad v = 1, 2, \dots, m. \quad (10)$$

Through this diffusion process, the SNF effectively suppresses weak edges (potential noise) while amplifying strong, consistent similarities present across modalities. This enables the method to fully leverage the local structure of each network and integrate both shared and complementary information, resulting in a robust consensus representation.

F. Deep Graph Learning

Based on the fused similarity matrix $P^{(\text{fused})}$, we use self-supervised graph learning to refine the multimodal graph and obtain cell embeddings. The fused graph is first sparsified by retaining the top- k neighbors for each cell and symmetrized. The resulting weighted graph is denoted as $\mathcal{G}^{(f)} = (\mathcal{V}, \mathcal{E}^+)$, where $\mathcal{E}^+$ contains positive cell-cell edges with weights $w_{uv} = P_{uv}^{(\text{fused})}$.

We implement the encoder as a two-layer graph convolutional network. The input node-feature matrix X was constructed by concatenating the low-dimensional scRNA-seq PCA representation and the scATAC-seq LSI representation after feature-wise standardization, thereby allowing the encoder to use modality-specific features while message passing is constrained by the SNF-fused graph. Given the normalized adjacency matrix $\tilde{A}$ and the input cell-feature matrix X , the hidden representation is computed as

$$H^{(1)} = \text{ReLU}(\tilde{A}X\Theta_0), \quad Z = \tilde{A}H^{(1)}\Theta_1, \quad (11)$$

where Θ_0 and Θ_1 are trainable parameters and $Z \in \mathbb{R}^{n \times d}$ is the learned embedding matrix. Dropout was applied to the hidden layer during training. Unless otherwise specified, the hidden dimension was set to 128 and the final embedding dimension was set to 64.

To train the encoder without additional labels, we randomly mask a subset $\mathcal{M}^+ \subset \mathcal{E}^+$ of positive edges and train the model to recover them from the remaining graph. To avoid a trivial solution in which all pairwise scores become high, we also sample a set of negative edges $\mathcal{M}^-$ from unconnected cell pairs with $|\mathcal{M}^-| = |\mathcal{M}^+|$. Negative edges were sampled from node pairs absent from the sparsified fused graph while excluding self-loops and duplicated pairs; the same sampling ratio was used across repeated runs. For an edge (u, v) , the predicted affinity is

$$s_{uv} = \sigma(\mathbf{z}_u^\top \mathbf{z}_v), \quad (12)$$

where $\sigma(\cdot)$ is the sigmoid function. The masked-edge reconstruction loss is defined as a weighted binary cross-entropy objective:

$$\mathcal{L}_{\text{mask}} = - \sum_{(u,v) \in \mathcal{M}^+} w_{uv} \log s_{uv} - \sum_{(u,v) \in \mathcal{M}^-} \log(1 - s_{uv}). \quad (13)$$

To preserve the observed graph structure, we further constrain the predicted affinities of non-masked observed edges $\mathcal{E}_{\text{obs}} = \mathcal{E}^+ \setminus \mathcal{M}^+$:

$$\mathcal{L}_{\text{obs}} = \sum_{(u,v) \in \mathcal{E}_{\text{obs}}} (s_{uv} - w_{uv})^2. \quad (14)$$

The full optimization objective is

$$\mathcal{L} = \mathcal{L}_{\text{mask}} + \alpha \mathcal{L}_{\text{obs}} + \beta \|\Theta\|_2^2, \quad (15)$$

where α balances masked-edge recovery and observed-edge preservation, β is the L_2 regularization coefficient, and Θ denotes all trainable parameters. In our implementation, $\alpha = 1$ and $\beta = 10^{-4}$ were used unless otherwise stated.

After training, the cell-specific embeddings Z_{cells} are used for clustering and visualization. When an integrated similarity matrix is required, we compute a bounded reconstructed network rather than a raw inner-product matrix:

$$W_{\text{integrated}}(u, v) = \sigma(\mathbf{z}_u^\top \mathbf{z}_v). \quad (16)$$

This transformation ensures that reconstructed cell-cell similarities are non-negative and lie in $[0, 1]$. For cell clustering, we applied the Leiden algorithm [35] to a nearest-neighbor graph built from Z_{cells} .

G. Community Detection via Leiden Algorithm

Given that the sxSNF framework relies on graph-based self-supervised learning, preserving the topological integrity of the learned latent structures is paramount during the clustering phase. Traditional methods like K-means assume convex cluster shapes, and the Louvain algorithm, while graph-based, often yields weakly connected or even disconnected communities. To address these limitations and fully leverage the high-fidelity graph topology captured by our embeddings, we employ the Leiden algorithm [36], which guarantees connectedness and provides a mathematically rigorous partition of the cellular landscape.

We first construct a nearest-neighbor graph from the learned embeddings Z_{cells} . Let this graph be denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents the set of cells. The clustering process aims to maximize the modularity quality function $\mathcal{Q}$, which quantifies the density of links inside communities compared to links between communities. We formulate the modularity optimization for a partition $\mathcal{P} = \{\mathcal{C}_1, \dots, \mathcal{C}_K\}$ as follows:

$$\mathcal{Q} = \frac{1}{2\Omega} \sum_{u,v \in \mathcal{V}} \left(\mathcal{A}_{uv} - \gamma \frac{d_u d_v}{2\Omega} \right) \delta(\sigma_u, \sigma_v) \quad (17)$$

Here, u and v denote arbitrary cell nodes in the graph. The term $\mathcal{A}_{uv}$ represents the affinity weight between cell u and cell v , and $\Omega = \frac{1}{2} \sum_{u,v} \mathcal{A}_{uv}$ signifies the total weight of all edges in the network. The variable $d_u = \sum_v \mathcal{A}_{uv}$ denotes the weighted degree of node u . The function $\delta(\sigma_u, \sigma_v)$ is the Kronecker delta, which equals 1 if cells u and v are assigned to the same community σ (i.e., $\sigma_u = \sigma_v$), and 0 otherwise. The resolution parameter γ controls the scale of cluster detection. To avoid symbol conflict with the regularization coefficient in the graph-learning objective, γ is used here for Leiden

resolution. Instead of fixing γ to a single value across all datasets, we selected it according to the expected biological granularity of each dataset and then kept it fixed for subsequent analyses. Specifically, we set $\gamma = 0.3$ for the PBMC-10k dataset, $\gamma = 0.5$ for the Chen-2019 dataset, and $\gamma = 0.4$ for the SHARE-seq dataset.

Unlike standard aggregation approaches, the Leiden algorithm explicitly checks for randomness in the local connectivity to partition the graph into sub-communities that are guaranteed to be connected. This is particularly crucial for our model, as the upstream self-supervised task (edge reconstruction) encodes subtle biological relationships into the graph structure.

H. Model Evaluation

To evaluate the performance of various methods, we use three widely used evaluation metrics including Adjusted Rand Index (ARI) [37], Normalized Mutual Information (NMI) [38], and Adjusted Mutual Information (AMI) [39]. ARI is used to measure the similarity between the model's predicted categories and the true categories, defined as:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}. \quad (18)$$

Here, a_i and b_j denote the numbers of cells in the i -th true cluster and the j -th predicted cluster, respectively, and n_{ij} denotes the number of cells shared by these two clusters. NMI is used to assess the consistency between predicted and true categories. Let $T = \{t_1, t_2, \dots, t_n\}$ and $Y = \{y_1, y_2, \dots, y_n\}$ denote the true and predicted categories of cells, respectively. NMI is defined as:

$$\text{NMI}(T, Y) = \frac{2\text{MI}(T, Y)}{H(T) + H(Y)} \quad (19)$$

The $H(\cdot)$ and $MI(\cdot)$ denote information entropy and mutual information, respectively. Considering that MI is higher if two clusters are of larger magnitude, regardless of whether there is more information to share, AMI adjusts the mutual information score, defined as:

$$\text{AMI}(T, Y) = \frac{\text{MI}(T, Y) - E(\text{MI}(T, Y))}{[H(T) + H(Y)]/2 - E(\text{MI}(T, Y))} \quad (20)$$

The $E(\cdot)$ denotes the mathematical expectation. The values of ARI and AMI are in the range of $[-1, 1]$, and the values of NMI are in the range of $[0, 1]$. The larger the values of these metrics, the better the performance of clustering.

In addition to clustering metrics, we evaluated integration quality using neighborhood-based batch mixing, modality mixing, and alignment score. All three metrics were computed on the k -nearest-neighbor graph of the learned embedding. For a cell i , let $N_k(i)$ denote its local neighbors. For a grouping variable g , the normalized mixing score was computed as

$$\text{Mix}(i; g) = \frac{\frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \mathbb{I}[g_j \neq g_i]}{1 - \sum_c p_c^2}, \quad (21)$$

where p_c is the global proportion of group c and the denominator is the expected different-group fraction under random

TABLE IV
SENSITIVITY ANALYSIS OF THE HYPERPARAMETERS (NEIGHBORHOOD SIZE k AND NUMBER OF ITERATIONS t), SHOWING THE EFFECT OF GRAPH CONSTRUCTION SETTINGS ACROSS REPEATED RUNS.

Neighborhood Size (k)	Number of Iterations (t)	ARI	NMI	AMI
10	10	0.542	0.718	0.715
10	20	0.602	0.821	0.819
20	10	0.623	0.724	0.722
20	20	0.694	0.833	0.832
30	20	0.556	0.726	0.724
50	20	0.651	0.723	0.721
20	30	0.555	0.825	0.823

mixing. Batch mixing used technical batch or sample labels as g , and modality mixing used assay labels as g . For alignment score, we computed, for each cell, the fraction of cross-modality neighbors that shared the same annotated cell type, averaged this value across all cells, and normalized it by the expected same-label fraction under random neighborhood assignment. Higher values indicate better mixing or alignment. These metrics were reported as mean $\pm$ standard deviation across repeated runs.

III. RESULTS

A. Performance comparison of multimodal integration methods

We first evaluated the quantitative performance of sxSNF on the PBMC-10k and SHARE-seq benchmark datasets. Chen-2019 was not included in the main benchmarking tables because it was primarily used for biological validation and interpretability analyses, including marker-gene validation, refined subpopulation analysis, and multimodal consistency assessment. The quantitative comparison on PBMC-10k and SHARE-seq is summarized in Table II, which reports the mean and standard deviation of ARI, NMI, and AMI across repeated runs. Overall, sxSNF achieved the best performance on both benchmark datasets across all three evaluation metrics, supporting its robustness across the evaluated datasets with distinct levels of multimodal sparsity and heterogeneity.

a) *Performance on PBMC-10k*: On the PBMC-10k dataset, sxSNF attains the best overall results in Table II, with an ARI of 0.694, NMI of 0.833, and AMI of 0.832. Compared with the strongest competing method, Seurat WNN, sxSNF improves ARI from 0.552 to 0.694 while also showing clear improvements in both NMI (0.833 vs. 0.726) and AMI (0.832 vs. 0.718). The gains over other recent deep-learning-based models are also consistent, including GLUE (ARI 0.583, NMI 0.702, AMI 0.695), MultiVI (ARI 0.488, NMI 0.695, AMI 0.688), and SIMBA (ARI 0.485, NMI 0.652, AMI 0.645). In contrast, conventional or weakly integrated methods such as Spectral clustering and scMIC remain far behind, confirming that robust multimodal fusion is critical for accurately resolving PBMC cell populations.

b) *Comparison across baseline methods*: Table II further shows a clear performance stratification across model classes. Early or shallow methods such as Spectral clustering and scMIC perform poorly on both datasets, whereas

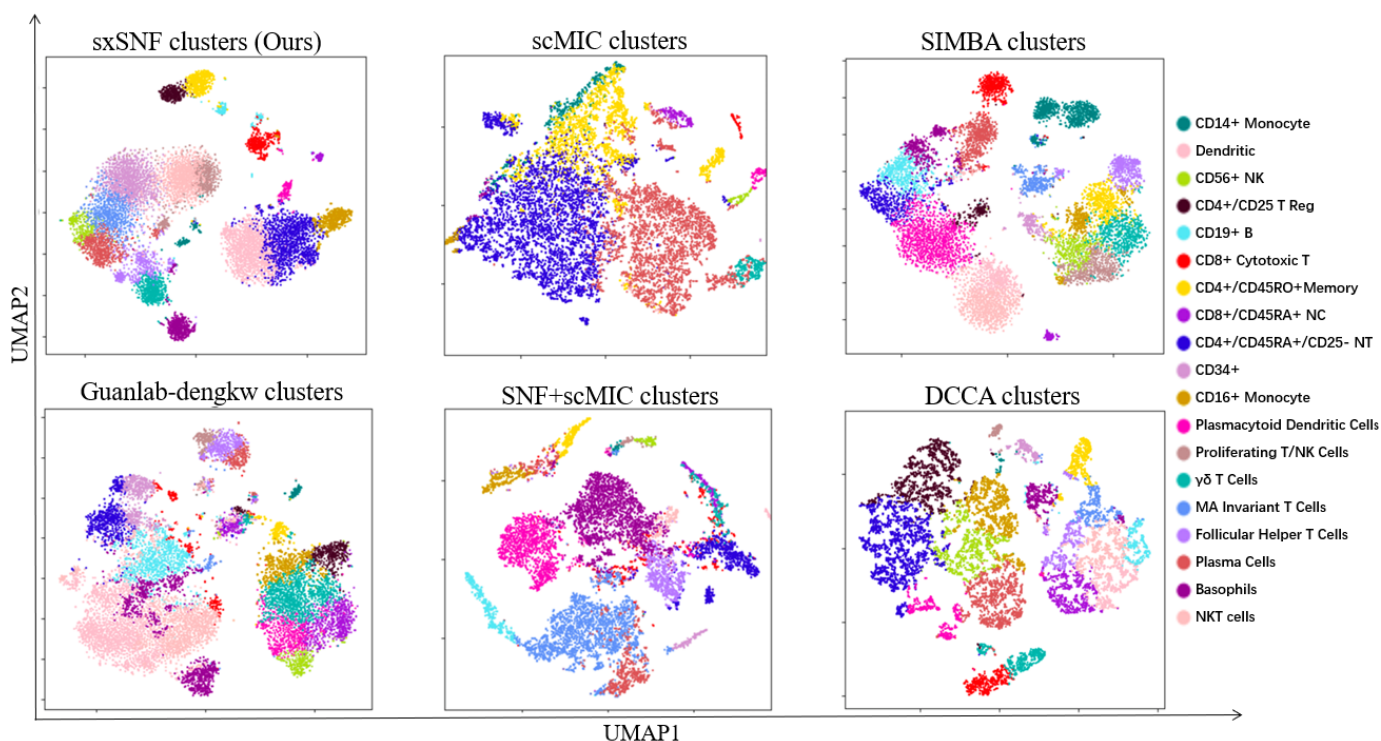


Fig. 2. Visualization of representative clustering results comparing single-cell multimodal integration methods on the PBMC-10k dataset. Representative methods are shown for visual comparison, and the full quantitative benchmark is provided in Table II.

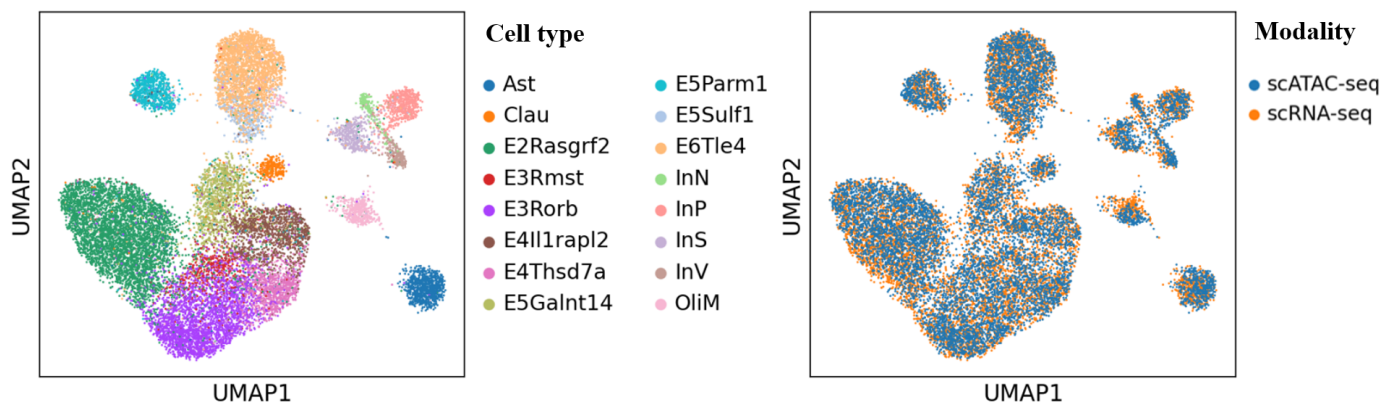


Fig. 3. UMAP plots of sxSNF fused scRNA-seq and scATAC-seq embedding from the Chen-2019 Dataset.

deep generative and graph-based integration methods, including MultiVI, GLUE, and Seurat WNN, achieve substantially stronger scores, with totalVI included as an auxiliary latent-variable reference. Nevertheless, sxSNF consistently exceeds these competitive baselines, suggesting that its fusion-before-learning strategy provides a more informative graph structure for downstream embedding. The relatively small standard deviations of sxSNF on both datasets further indicate that the method remains stable across repeated experiments rather than achieving isolated improvements in a few runs.

c) Performance on SHARE-seq: The same trend is even more evident on the more challenging SHARE-seq dataset. As reported in Table II, sxSNF reaches an ARI of 0.589, NMI of 0.768, and AMI of 0.761, outperforming Seurat WNN, the second-best method, on all three metrics (ARI 0.573, NMI

0.706, AMI 0.701). The margin is particularly notable for NMI and AMI, indicating that sxSNF more effectively captures the complex manifold structure and cell-type correspondence in data with stronger chromatin-accessibility heterogeneity. Compared with other strong baselines such as GLUE and MultiVI, as well as the auxiliary totalVI reference, the advantage of sxSNF becomes even clearer, further supporting the idea that iterative similarity fusion followed by deep graph representation learning is especially beneficial under highly sparse and heterogeneous multimodal settings.

d) Integration quality and cross-modal alignment: Beyond clustering accuracy, we evaluated how well the learned latent space mixed cells across batches and modalities using the metrics summarized in Table III. sxSNF achieved the best scores on both PBMC-10k and SHARE-seq for batch mixing,

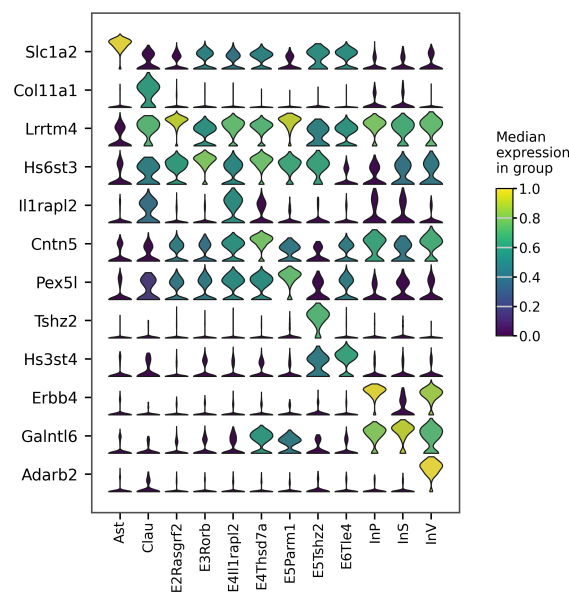


Fig. 4. Landscape of cell-type-specific marker genes in the Chen-2019 dataset visualized by a stacked violin plot.

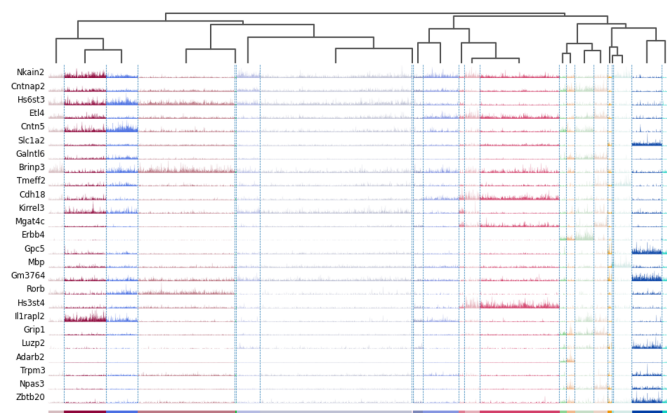


Fig. 5. Expression patterns of selected marker genes across various cell types. The heatmap displays gene expression levels (rows) across different cell clusters (columns), which are grouped by hierarchical clustering.

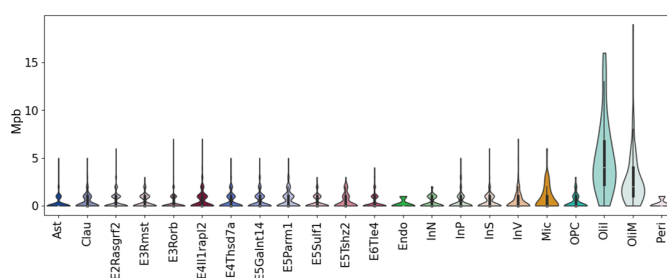


Fig. 6. Expression of *Mbp* (Oligodendrocyte Marker).

modality mixing, and overall alignment score, indicating that the fused graph representation improves not only label sepa-

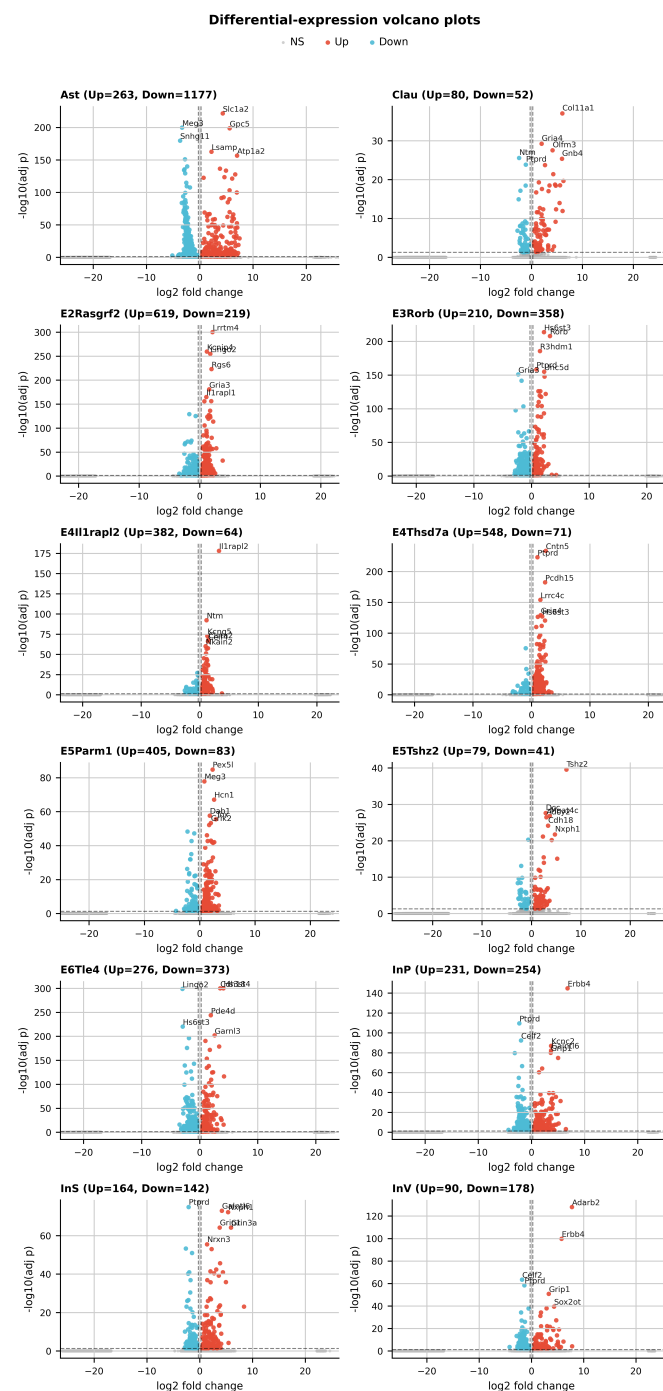


Fig. 7. Differential expression analysis supporting cluster identity validation in the Chen-2019 dataset. Each volcano plot shows cluster-specific genes for one annotated cell population; red and blue points indicate upregulated and downregulated genes, respectively, according to the differential-expression threshold used in the analysis.

ration but also integration consistency. On PBMC-10k, sxSNF increased the alignment score from 0.730 with GLUE to 0.795, while also improving batch mixing from 0.685 to 0.745 and modality mixing from 0.742 to 0.812. A similar advantage was observed on the more challenging SHARE-seq dataset, where sxSNF outperformed all competing methods with an alignment score of 0.768. These results suggest that the fusion-

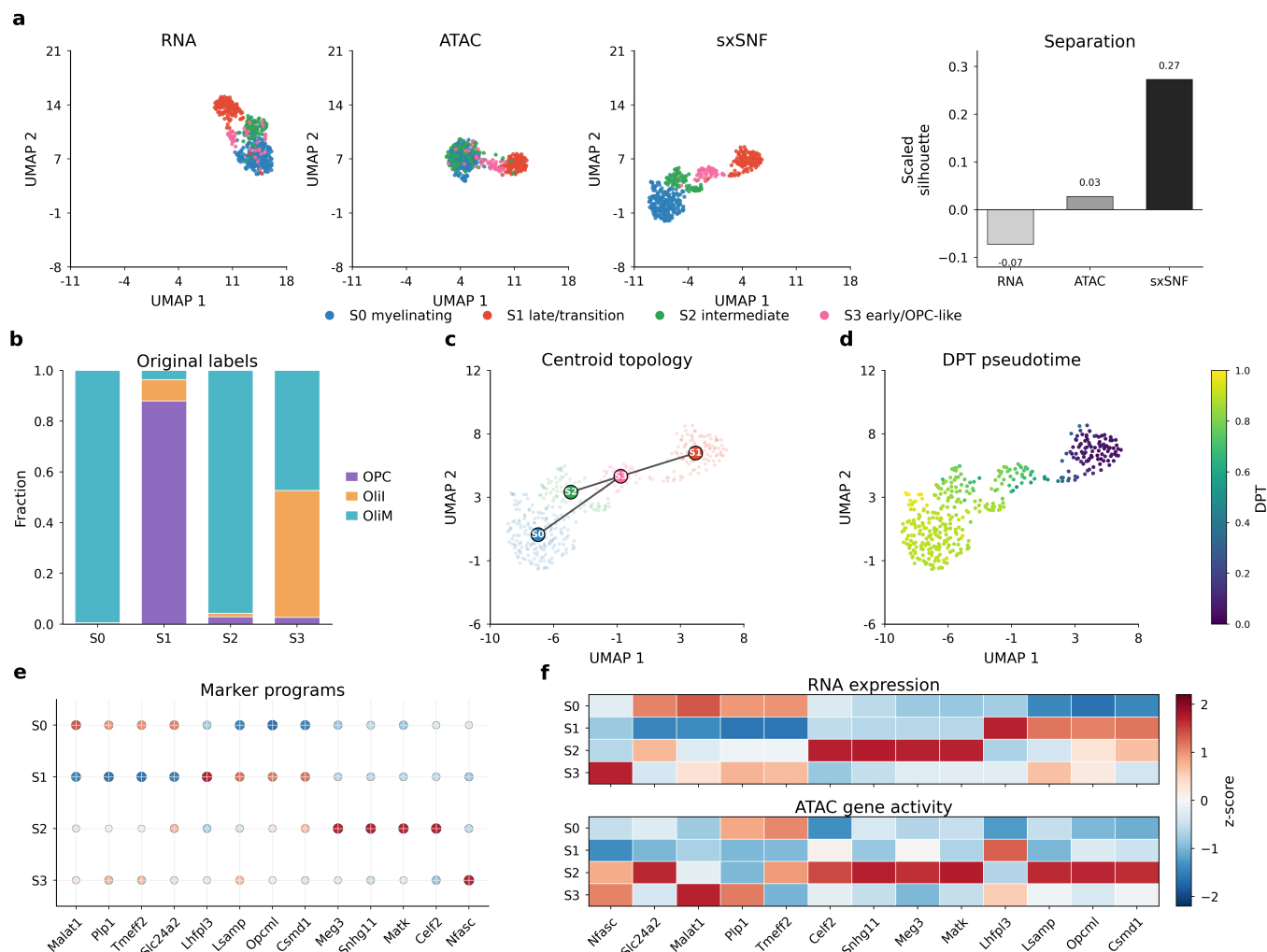


Fig. 8. Identification of refined oligodendrocyte-lineage subpopulations based on the integrated sxSNF embedding. (a) Comparison of RNA-only, ATAC-only, and sxSNF-integrated embeddings and corresponding subgroup-separation scores. (b) Original-label composition of refined subgroups. (c) Centroid-based subgroup topology. (d) DPT pseudotime ordering. (e) Subcluster marker programs. (f) Coordinated RNA expression and ATAC-derived gene activity across refined subgroups.

TABLE V

ABSOLUTE AND RELATIVE CHANGES IN CLUSTERING METRICS AFTER INCORPORATING SNF ON THE PBMC-10K BENCHMARK. RELATIVE GAINS ARE REPORTED AGAINST THE CORRESPONDING BASELINE IMPLEMENTATION.

Methods	Baseline			SNF-enhanced			Relative gain		
	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI
SIMBA	0.485	0.652	0.645	0.558	0.693	0.686	+15.04%	+6.36%	+6.42%
scMIC	0.114	0.301	0.291	0.352	0.607	0.593	+208.49%	+101.76%	+103.78%

before-learning strategy effectively reduces modality-specific bias while preserving biologically meaningful neighborhood structure, thereby producing a more coherent shared embedding for downstream analysis.

B. Synergistic benefits of SNF

As summarized in Table V, incorporating SNF improved the evaluated baseline integration algorithms across ARI, NMI, and AMI. The SNF+scMIC model showed the largest relative gain, with ARI increasing by 208.49% and NMI/AMI increasing by 101.76% and 103.78%, respectively. SNF+SIMBA also

showed measurable gains, with a 15.04% increase in ARI and approximately 6.4% improvements in NMI and AMI. Because relative gains can be inflated when the baseline score is low, these results are interpreted together with the absolute benchmark values in Table II. To visualize the structural changes brought by SNF, Fig. 9 compares cell-cell similarity networks before and after fusion. The initial modality-specific graphs, particularly the scATAC-seq graph, are sparse and fragmented; after SNF iteration, the consensus network increases local connectivity while preserving cell-type-aware neighborhood structure, as reflected by the neighbor-composition shift in

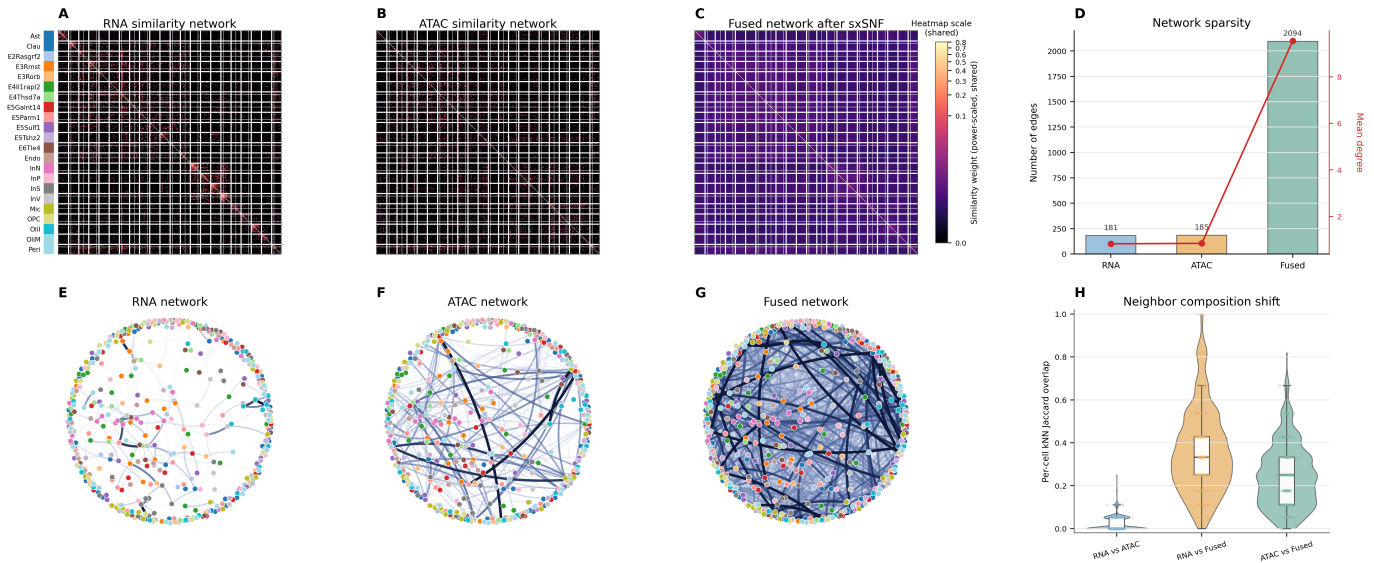


Fig. 9. Network-level interpretation of SNF fusion. The panels compare RNA-only, ATAC-only, and fused similarity networks using matrix and graph views under fixed visualization parameters, together with quantitative summaries of edge number, graph sparsity, and neighbor-composition shifts. The fused network increases within-neighborhood connectivity while preserving cell-type structure for downstream graph learning.

Fig. 9. This indicates that the fused network is not merely denser, but provides a more stable foundation for the downstream graph-learning module.

C. Ablation study and component contribution

To rigorously evaluate the contribution of each core component within the sxSNF framework—specifically the Similarity Network Fusion (SNF) mechanism and the Self-Supervised Graph Learning (SSGL) module—we designed a comprehensive ablation study. We compared the full sxSNF model against three variants:

- **Variant A (w/o Integration):** This baseline performs clustering separately on the scRNA-seq and scATAC-seq modalities using standard PCA and Louvain clustering, representing the performance without any multi-view fusion.
- **Variant B (w/o Iterative Fusion):** In this variant, the iterative SNF cross-diffusion updates are removed. Instead, the similarity matrices from both modalities are simply averaged ($P = 0.5(P^{(\text{scRNA})} + P^{(\text{scATAC})})$) to serve as the input for the downstream graph learning, testing the necessity of the message-passing mechanism in SNF.
- **Variant C (w/o Self-Supervised Learning):** This variant retains the SNF fusion but replaces the deep graph learning module with standard Spectral Clustering on the fused matrix $P^{(\text{fused})}$. This evaluates the advantage of the edge-reconstruction pretext task over traditional decomposition methods.

The ablation results on the PBMC-10k dataset are summarized in Table VI.

Variant A shows that either modality alone is insufficient to recover the full cell-type structure. Variant B improves over the single-modality baselines, indicating that combining modalities is beneficial, but simple averaging remains weaker

TABLE VI
ABLATION STUDY RESULTS ON PBMC-10K DATASET: CONTRIBUTION OF SNF AND SELF-SUPERVISED LEARNING COMPONENTS

Method / Variant	Components	ARI	NMI	AMI
Variant A (scRNA only)	-	0.3852	0.5120	0.5089
Variant A (scATAC only)	-	0.3241	0.4635	0.4601
Variant B (Average Fusion)	SSGL only	0.4933	0.6450	0.6412
Variant C (SNF + Spectral)	SNF only	0.5105	0.6682	0.6645
sxSNF (Full Model)	SNF + SSGL	0.6940	0.8330	0.8320

Note: "SSGL" denotes the Self-Supervised Graph Learning module. All ablation variants were evaluated on the same PBMC-10k benchmark setting used in Table II. Variant B uses simple averaging instead of SNF iterative fusion. Variant C uses Spectral Clustering instead of the deep learning embedding.

than iterative SNF. Variant C further improves the results by using the SNF-fused graph, showing that the fusion step itself contributes substantially to graph quality. The full sxSNF model achieves the highest ARI, NMI, and AMI, indicating that SNF-based graph denoising and self-supervised graph learning provide complementary gains. The relatively large gap between Variant C and the full model further suggests that the masked-edge reconstruction objective does not merely project the fused graph, but further reshapes the fused topology into a clustering-favorable latent space.

D. Visualization and biological interpretability of integrated embeddings

To visually assess clustering quality and multimodal alignment, we applied uniform manifold approximation and projection (UMAP [28]) to latent representations derived from each integration method. In Fig. 2, embeddings are colored by ground-truth cell-type labels. The sxSNF method produced well-separated clusters with sharp boundaries, closely matching the underlying biological structure.

We further evaluated the joint embedding of scRNA-seq and scATAC-seq data from the Chen-2019 dataset (Fig. 3). The left panel, colored by cell type, reveals 18 distinct clusters that correspond to known biological identities. The right panel, colored by assay modality (scATAC-seq in blue, scRNA-seq in orange), shows strong interleaving of both modalities within each cluster. This indicates successful removal of technique-specific bias while preserving discriminative cell-type structure in a unified latent space.

E. Marker gene expression supports cluster annotations

To validate the cellular identities obtained from the Chen-2019 SNARE-seq dataset, we examined marker-gene expression patterns across annotated cortical populations. The observed distribution of canonical markers closely recapitulated established cortical cell-type taxonomy, indicating robust cluster assignment and high data integrity (Fig. 4). To further support cluster identity validation beyond qualitative marker visualization, we performed differential expression (DE) analysis for each annotated cell population. The resulting volcano plots show clear cluster-specific transcriptional signatures, with strongly upregulated genes corresponding to the expected biological identities of major neuronal and glial populations (Fig. 7).

Astrocytes (Ast) exhibited prominent expression of *Gpc5*, a glypican-related marker associated with glial transcriptional programs. Mature oligodendrocytes (OliM) showed strong expression of *Plp1*, a canonical myelin-associated gene and well-established oligodendrocyte lineage marker. These glial signatures are consistent with the expected cell identities recovered from the Chen-2019 cortical dataset.

Among excitatory neurons, transcriptomic profiles resolved into layer-associated subtypes. Cells annotated as E3 Rorb expressed high levels of *Rorb*, whereas E4 Il1rapl2 populations were enriched for *Il1rapl2*. These patterns are consistent with the laminar organization of cortical excitatory neurons and support the biological interpretability of the sxSNF embedding.

Inhibitory interneuron groups (InN, InP, InS, and InV) displayed elevated *Adarb2*, supporting their interneuron-related transcriptional identity. Additional neuronal markers demonstrated coherent cell-class specificity: *Kcnq5* was enriched broadly across neuronal populations but absent in glial populations, and synaptic adhesion genes such as *Nxph1* were preferentially expressed in neuronal clusters, including interneuron subsets. Together, these marker-gene and DE results indicate that sxSNF-derived embeddings preserve biologically meaningful transcriptional programs and support reliable cell-type annotation in the Chen-2019 validation dataset.

F. Case Study: Transcriptional Heterogeneity and Marker Validation

To characterize global transcriptional heterogeneity, we visualized expression profiles of top-ranked genes across annotated clusters. Hierarchical clustering, as shown in the tracksplot dendrogram, successfully reconstructed cellular lineage relationships, clearly separating neuronal subtypes (e.g.,

excitatory *E3/E4* populations) from non-neuronal glial populations (e.g., *Ast* and *OliM*) (Fig. 5). The resulting expression landscape exhibits distinct specificity, with marker genes showing dense, high-magnitude expression peaks restricted to corresponding cell types. This block-like expression pattern indicates that the clustering algorithm successfully resolved biologically distinct populations based on transcriptomic signatures, rather than technical noise.

For quantitative validation of cell-type identity, we performed a focused analysis on *Mbp* (Myelin Basic Protein) (Fig. 6), a canonical marker essential for structural integrity of the central nervous system myelin sheath [40]. The distribution of *Mbp* expression was visualized using violin plots without Y-axis truncation to preserve the full dynamic range of expression data. Results reveal a highly specific expression pattern: the Mature Oligodendrocyte (*OliM*) cluster displays an exclusive, high-magnitude expression profile, whereas Oligodendrocyte Precursor Cells (*OPC*) and neuronal clusters show negligible expression levels. This distinct stratification aligns with established spatial transcriptomic atlases of the hypothalamic preoptic region [41], confirming that computational annotation of the *OliM* cluster accurately reflects its biological phenotype.

G. Robustness and Hyperparameter Sensitivity Analysis

Since single-cell integration can be sensitive to random initialization and graph construction parameters, we systematically evaluated the robustness of sxSNF. Specifically, the choice of neighborhood size k dictates the extent of local structural preservation: too small a k risks fragmenting the graph, while too large a k may introduce noise from dissimilar cells. Similarly, the number of diffusion iterations t controls the depth of cross-modal information propagation: insufficient iterations lead to inadequate fusion, whereas excessive iterations can cause over-smoothing. To justify our chosen settings (defaulting to $k = 20, t = 20$), we performed repeated runs across different random seeds and varied these key SNF hyperparameters.

As detailed in the sensitivity analysis (Table IV), sxSNF performance depends on the joint choice of neighborhood size k and diffusion iteration number t . The best clustering performance was obtained at $k = 20$ and $t = 20$, supporting this setting as the default configuration used in the main benchmark experiments. Alternative settings produced lower but internally stable performance across repeated runs, indicating that graph construction parameters should be selected carefully while the model remains functional across a broad range of tested values.

H. Refinement of Cell Subpopulations and Subgroup Relationships

To assess whether sxSNF provides biological insight beyond recapitulating known marker-gene patterns, we performed a lineage-restricted analysis of the oligodendrocyte lineage in the Chen-2019 dataset, including oligodendrocyte precursor cells (OPC), intermediate oligodendrocytes, and mature oligodendrocytes. As illustrated in Fig. 8, reclustering within the sxSNF-integrated embedding resolved four refined subgroups

that were more clearly separated than in the RNA-only or ATAC-only representations, which was further supported by the highest scaled silhouette score in the sxSNF space. These subgroups exhibited distinct original-label compositions, including mature oligodendrocyte-enriched, OPC-enriched, and intermediate/transitional states. A centroid-based topology graph together with DPT pseudotime analysis further showed an organization consistent with oligodendrocyte maturation.

Subcluster-specific marker analysis identified a mature myelinating subgroup enriched for *Plp1*, *Mbp*, *Mag*, *Mog*, *Mobp*, and *Aspa*, whereas the remaining subgroups displayed distinct regulatory or transitional gene programs. Together, these results are consistent with an ordered oligodendrocyte maturation-related structure from early/OPC-like states to mature myelinating states, rather than simply partitioning cells according to the original coarse annotations. In addition, subgroup-averaged RNA expression and proximal ATAC-derived gene activity showed coordinated subgroup-specific patterns, indicating that sxSNF captures refined, functionally interpretable, and multimodally supported cellular states rather than merely reproducing known marker-gene signatures.

IV. DISCUSSION AND CONCLUSION

In this study, we present sxSNF, a multimodal single-cell integration framework that couples Similarity Network Fusion (SNF) with self-supervised graph learning. Instead of applying graph learning directly to a potentially noisy single-modality graph, sxSNF first performs iterative cross-modal diffusion to construct a consensus graph and then uses a masked-edge reconstruction task to learn cell embeddings from this denoised structure. This fusion-before-learning design provides an explicit structural prior for graph representation learning and helps reduce the impact of modality-specific sparsity, particularly in scATAC-seq data.

The two core components of sxSNF play complementary roles. SNF acts as a structural consensus-building step by reinforcing local neighborhoods supported across modalities and reducing weak modality-specific edges. The graph-learning module then maps the fused topology into a compact latent space and refines connectivity through self-supervised reconstruction. The ablation results suggest that neither component alone fully explains the final performance: simple averaging is weaker than iterative SNF, and SNF followed by spectral clustering is weaker than the complete self-supervised sxSNF model.

The benchmark experiments showed that sxSNF achieved higher clustering scores than the evaluated baselines on PBMC-10k and SHARE-seq. The integration metrics further suggest improved batch and modality mixing in the learned embedding. On the Chen-2019 SNARE-seq dataset, the sxSNF embedding recovered known cortical cell populations, marker-gene programs, and refined oligodendrocyte-lineage states supported by both RNA expression and ATAC-derived gene activity. These results indicate that the learned representation preserves biologically meaningful structure rather than only optimizing clustering scores.

Several limitations should be noted. First, the present study focuses mainly on RNA+ATAC integration; additional valida-

tion is needed for CITE-seq, spatial multi-omics, and other modality combinations. Second, sxSNF still depends on the quality of the initial nearest-neighbor graphs, and very weak preprocessing or severe batch imbalance may affect the fused graph. Third, Leiden resolution influences the final number of clusters, although we fixed the resolution within each dataset after selecting the expected biological granularity. Fourth, the sensitivity analysis indicates that the joint choice of neighborhood size and fusion depth can influence clustering accuracy, suggesting that adaptive or data-driven parameter selection will be important in future work. Fifth, the million-cell scalability analysis relies partly on ANN graph construction and sparse computation assumptions; larger empirical benchmarks will be needed to fully evaluate performance on atlas-scale datasets.

Future work will explore adaptive modality weights, end-to-end coupling between graph fusion and representation learning, and downstream applications such as trajectory inference, perturbation response modeling, and cell-cell communication analysis. Overall, sxSNF provides a graph-centered framework for multimodal single-cell integration by combining SNF-based structural refinement with self-supervised representation learning.

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